

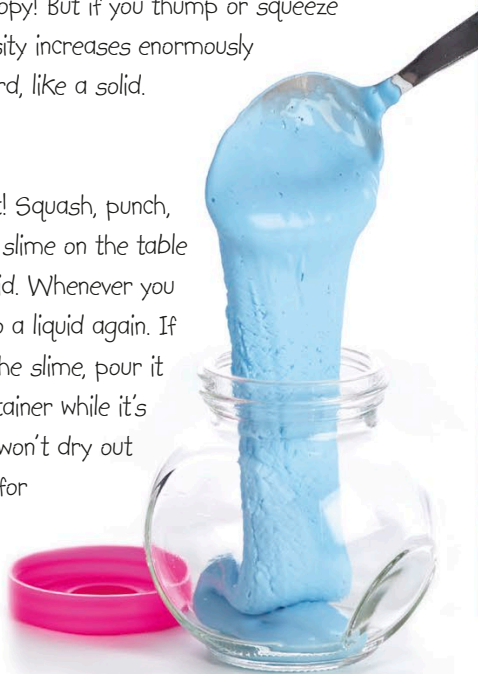


3 Add a few tablespoonfuls of warm water. Keep stirring to mix the water into the cornstarch. The water makes starch (a substance in the cornstarch) expand, forming a network that holds the water and cornstarch together in a slimy mixture.



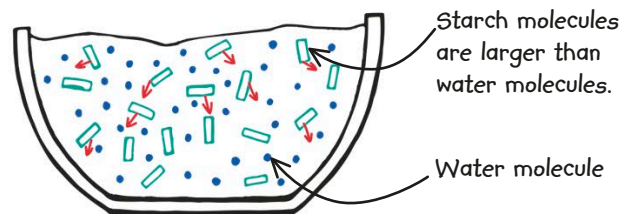
4 Gradually, your mixture will turn into a thick paste. Pick it up and knead it in your hands—it will get really gloopy! But if you thump or squeeze the slime, its viscosity increases enormously and it becomes hard, like a solid.

5 Now go for it! Squash, punch, or slam your slime on the table to make it turn solid. Whenever you stop, it will turn into a liquid again. If you want to keep the slime, pour it into an airtight container while it's runny. That way it won't dry out and you can use it for about a month.



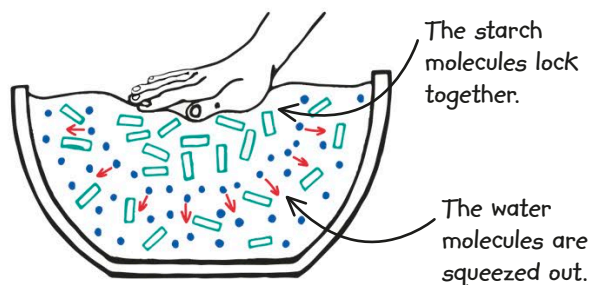
HOW IT WORKS

A molecule is the smallest part of a compound. It's the starch molecules reacting with water that are responsible for the slime's viscosity. As long as the molecules can move around, the mixture stays liquid. Sudden pressure, though, makes the molecules jam together, so the mixture can't flow.



WITHOUT PRESSURE

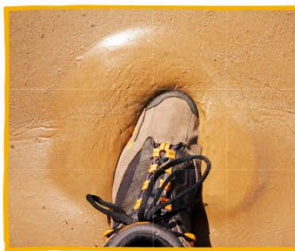
As long as you handle the slime gently and don't squish it too hard, the starch molecules can move around, suspended in the water. This makes a thick, slow-flowing liquid.



WITH PRESSURE

If you press hard on the slime, you squeeze out the water molecules from between the starch molecules, which lock together and make the slime feel more solid.

REAL WORLD SCIENCE QUICKSAND



A liquid with a viscosity that changes under pressure is known as a non-Newtonian fluid. Some of these liquids, such as slime, get thicker and behave like solids. But quicksand—a mixture of sand, clay, and water—is an example of a liquid that gets runnier. If you get stuck in quicksand and struggle to get out, your movements will cause you to sink.